

Customer Retention & Churn Analysis Report

Project Overview

Project Title: Task 2 – Customer Retention & Churn Analysis

Internship Program: Data Science & Analytics Internship 2026

Tools Used: Python (Pandas, NumPy, Matplotlib, Seaborn), Power BI, Excel

Objective: Analyze subscription customer data to identify churn patterns, retention drivers, and actionable strategies to reduce customer loss.

Executive Summary

This project analyzes customer behavior in a subscription-based business using a churn dataset similar to the IBM Telco Customer Churn dataset. The analysis identifies the most significant factors influencing churn, including contract type, tenure, monthly charges, and support service adoption. Key findings show that month-to-month customers, new customers, and customers with higher monthly charges are substantially more likely to churn. Recommendations include promoting long-term contracts, improving onboarding, and targeting at-risk segments with personalized offers.

Dataset Description

The dataset contains customer-level information such as:

- Demographics (gender, senior citizen, partner, dependents)
- Account details (tenure, contract type, payment method)
- Services subscribed (internet, security, tech support)
- Financial data (monthly charges, total charges)
- Target variable: **Churn** (Yes/No)

Data Cleaning & Preparation

The following preprocessing steps were performed:

- Removed duplicate records
- Converted TotalCharges to numeric
- Handled missing values
- Standardized categorical values
- Encoded the target column (Churn)
- Created tenure groups for cohort analysis

Exploratory Data Analysis (EDA)

Major EDA tasks included:

- Overall churn rate calculation
- Churn by contract type

- Churn by payment method
- Churn by tenure group
- Monthly charges comparison
- Service-wise churn analysis

Key Findings

- 1. Overall Churn Rate:** Approximately 26–27% of customers churned.
- 2. Contract Type:** Month-to-month customers showed the highest churn rate, while two-year contract customers had the lowest.
- 3. Tenure:** Customers with less than 12 months of tenure churned the most.
- 4. Monthly Charges:** Customers paying higher monthly charges were more likely to leave.
- 5. Tech Support & Security:** Customers without online security and tech support churned at a significantly higher rate.
- 6. Payment Method:** Electronic check users exhibited elevated churn.

Customer Lifetime Value (CLV) Insights

Estimated CLV was calculated as:

CLV = Average Monthly Revenue × Average Customer Lifetime

Long-tenure customers on annual contracts generated the highest lifetime value, reinforcing the importance of retention strategies focused on these segments.

Cohort Retention Analysis

Customers were grouped by signup month and tracked across subsequent months. Retention curves showed strong drop-off during the first three months, after which retention stabilized. This indicates that onboarding and early engagement are critical to long-term retention.

Business Recommendations

- Offer discounts to encourage one-year and two-year contracts.
- Build a structured onboarding journey for the first 90 days.
- Promote online security and tech support add-ons.
- Target high monthly charge customers with loyalty incentives.
- Monitor churn risk scores and proactively contact at-risk customers.
- Improve payment experience for electronic check users.

Suggested Dashboard Components

- KPI Cards: Total Customers, Churn Rate, Average Tenure, Average CLV
- Bar Chart: Churn by Contract Type
- Heatmap: Correlation Matrix
- Cohort Retention Matrix
- Box Plot: Monthly Charges vs Churn

- Segment Filters: Contract, Internet Service, Payment Method

Python Code Workflow

1. Load dataset using Pandas.
2. Clean and preprocess data.
3. Perform EDA and churn analysis.
4. Build cohort retention table.
5. Estimate customer lifetime value.
6. Export charts and create dashboard/report.

GitHub Repository Structure

FUTURE_DS_02/

■■■ data/customer_churn.csv

■■■ notebooks/churn_analysis.ipynb

■■■ dashboard/churn_dashboard.pbix

■■■ reports/Customer_Retention_and_Churn_Analysis_Report.pdf

■■■ README.md

Conclusion

Customer churn is concentrated among new customers on month-to-month contracts with higher monthly charges and limited support services. By focusing on onboarding, contract upgrades, and proactive retention campaigns, businesses can significantly improve customer lifetime value and reduce revenue loss.